

3.9 H: Integration

	Content
H1	Know and use the Fundamental Theorem of Calculus.

	Content
H2	Integrate x^n (excluding $n = -1$), and related sums, differences and constant multiples. Integrate e^{kx}, $\frac{1}{x}$, $\sin kx$, $\cos kx$ and related sums, differences and constant multiples.

	Content
H3	Evaluate definite integrals; use a definite integral to find the area under a curve and the area between two curves.

	Content
H4	Understand and use integration as the limit of a sum.

	Content
H5	Carry out simple cases of integration by substitution and integration by parts; understand these methods as the inverse processes of the chain and product rules respectively. (Integration by substitution includes finding a suitable substitution and is limited to cases where one substitution will lead to a function which can be integrated; integration by parts includes more than one application of the method but excludes reduction formulae).

	Content
H6	Integrate using partial fractions that are linear in the denominator.

H7	Evaluate the analytical solution of simple first order differential equations with separable variables, including finding particular solutions (Separation of variables may require factorisation involving a common factor).
	Content
H8	Interpret the solution of a differential equation in the context of solving a problem, including identifying limitations of the solution; includes links to kinematics.